SharePoint metadata tagging pipeline

Proposed Azure implementation – initial backfill and future scheduled runs
Only blank target fields are updated

Key

- Primary flow
- Supporting flow
- No clear rule flow (to AI)
- Authorisation flow
- Graph connection / data flow

1 Development and deployment

VS Code
Python rules and tests

Azure DevOps
Repository and deployment

Deploy

2 Client Azure environment + Connected Microsoft Graph service

Deployed in client Azure

Read files and metadata

Azure Function
Timer and file discovery
- Backfill + future runs
- Reads files and metadata

Python rules engine
- Approved Excel rules
- Only missing fields

Rule match

Validate and recheck
- Approved values only
- Preserve populated fields

Validated writes

No clear rule

Pipeline identity
- Entra app or managed identity
- Site grant

Azure Storage
- Function host, optional
- processing checkpoints

Application Insights
- Run counts and errors
- No document content in logs

Authorise calls

Connected service

Microsoft Graph
SharePoint read / metadata write
Microsoft-hosted API (not deployed in the client Azure resource group)

3 External dependencies and decisions

Graph connection
(reads test files)

SharePoint test library
- Files, metadata, approved tag list
- Read-only for testing

Approved AI service
Only if the rules cannot decide

Uncertain

Human review
For unresolved or uncertain cases

Testing: read-only first; approved metadata write on named test files only.
Key Vault is optional if secret-based credentials are required.